

ACE-CAN attack-centric CAN IDS pipeline

Repair the objective first, then test feature-preserving detection.

Input conditions

I1 Rare traces

CAN ID, DLC, payload
timestamp, source
attack rate < 1%

I2 Cross-shift

unknown vehicle
unknown attack family
normal traffic dominates

Branch operation

E1 Metric audit

all-normal check
weighted-score ambiguity
ranking inversion

D1 GRAIN-CAN

same-ID timing
payload dynamics
ID behavior

Evidence outputs

E2 Corrected eval

attack P/R/F1
AUPR, R@FPR
event recall

D2 Claim boundary

strong baseline
low-FPR evidence
not solved

metric trap:
high score, zero recall

Attack-F1 **AUPR** **R@FPR** **Event**

Output: corrected benchmark + deployment evidence + reproducible reporting checklist